

Auren Ng

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EDUCATION

MSc Sensor Systems Engineering, Imperial College London 2025 - 2026

- Reproduced federated learning experiments from published smart-wheelchair navigation research using MATLAB, benchmarking different distributed optimization methods
- Designed a high bandwidth (10kHz – 100 kHz) lock-in with phase-sensitive detection techniques using low-noise op-amps STM32 controller
- Designed an environmental sensor module by combining a gas sensor node, GPS, and time-of-flight sensor, controlled by a STM32 microcontroller firmware implemented in C

BEng Mechatronic Engineering (First Class), The University of Manchester 2022 - 2025

- Awarded £3000 per year as a Power Academy scholar under the IET
- Programmed firmware for an STM32 device in C to function as a smart clock using interrupts, digital debouncing, and efficient state machine design

WORK EXPERIENCE

Electronics Engineering Intern, UK Atomic Energy Authority June – Sept 2024

- Interconnect PCB design using Proteus to monitor and relay signals between IGBT driver stack and 3-phase power delivery for nuclear reactor's filament heating system
- Designed digital logic circuit for IGBT driver signal fault monitoring allowing microcontroller to flag faults under defined triggers, i.e. over-temperature or over-current conditions

PROJECTS

Intelligent Pathfinding for a Powered Wheelchair through Crowded Environments 2025 - 2026

- C programming using ROS2 libraries on Ubuntu to feed motor commands through system
- Low-level communication protocol implementation between CAN (on Linux) and RNet
- Integrated ZED X camera and LiDAR data using ROS2 architecture for real-time perception

Digital Systems Laboratory in Xilinx ISE 2024 - 2025

- Designed and verified digital hardware components, i.e. finite state machines (FSM) and sequential logic circuits, in VHDL using assertion-based verification
- Developed VHDL testbenches to verify clocking, reset behaviour, state transitions, and counter operations of synchronous digital systems through simulation
- Implemented and verified FSM digital designs in VHDL, integrating existing IP cores and analysing post-synthesis timing behaviour using Xilinx ISE simulation tools
- Black-box testing and reverse modelling of hardware IP cores in VHDL through simulation-driven analysis of state-machine behaviour and timing characteristics

Magnetic Levitating Platform with Model-based Control 2024 - 2025

- Prototyping in MATLAB for magnetic levitator by modelling magnetic field
- Simulink modelling for LQR controller to analyse convergence rate and boundary conditions
- Implementation of a discrete-time controller on Arduino Uno to power solenoid array based on target position feedback from Hall-effect sensor array

Embedded Systems Project – Line-following Buggy 2023 - 2024

- C++ programming on STM32 to implement cascaded PID loop to control heading and speed
- Embedded system design including Bluetooth to feedback control parameters in real time
- PCB design in Altium Designer to produce infrared sensor array for line detection